

Printable Assignment - Class: PHYS 2100 - Fall 2024 - Malikova Assignment: Chapter 21

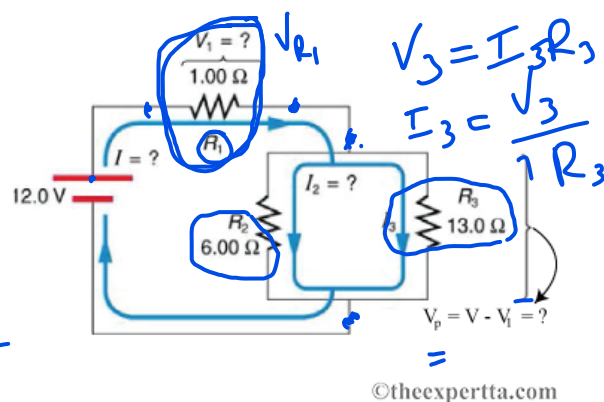
Problem 1: As shown, three resistors are connected to a voltage source such that the resistors R_2 and R_3 are in parallel with one another, and that parallel combination is in series with resistor R_1 and the voltage source. Resistance values and the potential of the voltage source are as indicated in the figure.

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$$R_{eq} = 1 + \frac{1}{\frac{1}{13} + \frac{1}{6}} = 97/19 \Omega = 5.105 \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{12}{5.105} = 2.35 \text{ A}$$

$$I_3 = \frac{9.65}{13}$$



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Calculate the current, I_3 , in amperes, that passes through resistor R_3 .

Numeric : A numeric value is expected and not an expression.

$I_3 =$ _____ A

0.742 A

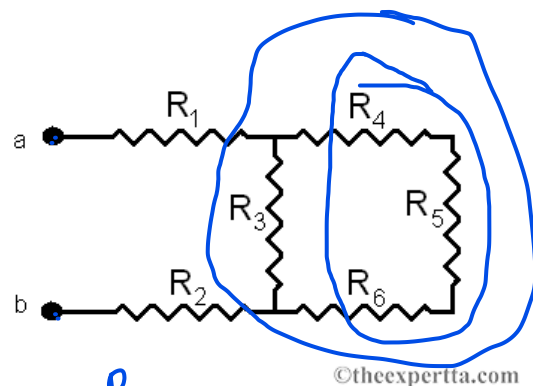
$$\frac{1}{\frac{1}{13} + \frac{1}{6}} = 4.105 \Omega$$

$$4.105 \times 2.35 = 9.647 \text{ V}$$

Problem 2: A circuit made up of 6 resistors is shown in the figure, with resistances $R_1 = 25 \Omega$, $R_2 = 65 \Omega$, $R_3 = 25 \Omega$, $R_4 = 65 \Omega$, $R_5 = 87 \Omega$, and $R_6 = 64 \Omega$. The total current going through the circuit is $I = 16 \text{ A}$.

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$$\Delta V = I R_{eq}$$



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Part (a) Express the equivalent resistance of the combination of R_4 , R_5 , and R_6 .

Expression :

$R_{eq} =$ _____

$$R_4 + R_5 + R_6$$

Select from the variables below to write your expression. Note that all variables may not be required.

$\alpha, \beta, \theta, d, g, I, m, R, R_1, R_2, R_3, R_4, R_5, R_6, t$

Part (b) Express the equivalent resistance of the combination of R_3 , R_4 , R_5 and R_6 .

Expression :

$R'_{eq} =$ _____

$$\frac{1}{\left(\frac{1}{R_3} + \frac{1}{R_4 + R_5 + R_6}\right)}$$

Select from the variables below to write your expression. Note that all variables may not be required.

$\alpha, \beta, \theta, d, g, I, m, R, R_1, R_2, R_3, R_4, R_5, R_6, t$

Part (c) Express the equivalent resistance R of the combination of R_1 , R_2 , R_3 , R_4 , R_5 and R_6 .

Expression :

$R =$ _____

$$R_1 + R_2 + \frac{1}{\frac{1}{R_3} + \frac{1}{R_4 + R_5 + R_6}}$$

Select from the variables below to write your expression. Note that all variables may not be required.

$\alpha, \beta, \theta, d, g, I, m, R, R_1, R_2, R_3, R_4, R_5, R_6, t$

Part (d) Calculate the numerical value of the equivalent resistance R in Ω .

Numeric : A numeric value is expected and not an expression.

$R =$ _____ /

Part (e) Express the potential difference between a and b, ΔV , in terms of the total current I and the equivalent resistance of the entire circuit R .

Expression :

$\Delta V =$ _____ $I R_{eq}$

Select from the variables below to write your expression. Note that all variables may not be required.

$\alpha, \beta, \theta, d, g, I, m, R, R_1, R_2, R_3, R_4, R_5, R_6, t$

Part (f) Calculate the numerical value of ΔV in V.

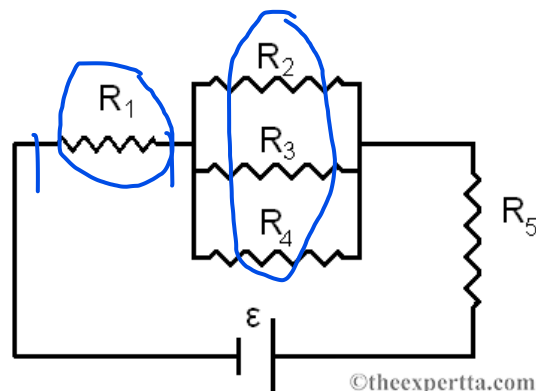
Numeric : A numeric value is expected and not an expression.

$\Delta V =$ _____ /

Problem 3: A circuit consisting of 5 resistors is shown in the graph. Their resistances are $R_1 = 19 \Omega$, $R_2 = 15 \Omega$, $R_3 = 89 \Omega$, $R_4 = 74 \Omega$, and $R_5 = 25 \Omega$, and the emf of the battery is $\varepsilon = 2.5 \text{ V}$. Suppose the internal resistance of the battery is zero.

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$$\varepsilon = I R_{eq}$$



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Part (a) Express the equivalent resistance of the combination of R_2 , R_3 , and R_4 .

Expression :

$R_{eq} =$ _____

$$1 / \left(\frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \right)$$

Select from the variables below to write your expression. Note that all variables may not be required.

R_2, R_3, R_4

Part (b) Express the total resistance of the circuit R in terms of R_1 , R_2 , R_3 , R_4 , and R_5 .

Expression :

$R =$ _____ $R_1 + R_5 + 1 / \left(\frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \right)$

Select from the variables below to write your expression. Note that all variables may not be required.

R_1, R_2, R_3, R_4, R_5

Part (c) Calculate the numerical value of the total resistance R in ohms.

Numeric : A numeric value is expected and not an expression.

$R =$ _____ Ω

Part (d) Express the current I through R_1 in terms of the emf ε and the equivalent resistance R .

Expression :

$I =$ _____

Select from the variables below to write your expression. Note that all variables may not be required.

ε, R

Part (e) Calculate the numerical value of I in amps.

Numeric : A numeric value is expected and not an expression.

$I =$ _____ A

$$P = IV = \frac{\varepsilon^2}{R} = I^2 R$$

Part (f) Express the power P dissipated by R_1 using I and R_1 .

Expression :

$$P = I^2 R_1$$

Select from the variables below to write your expression. Note that all variables may not be required.

I, R_1

Part (g) Calculate the numerical value of the power dissipated by R_1, P , in watts.

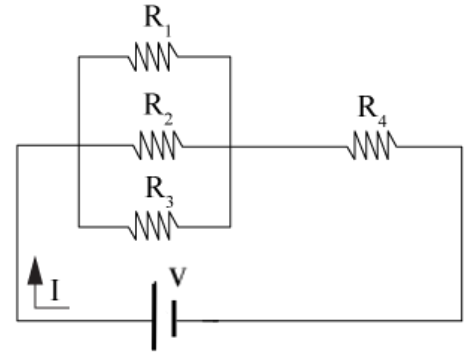
Numeric : A numeric value is expected and not an expression.

$$P = \text{_____} \text{ W}$$

Problem 4: Consider the circuit in the following diagram, where the resistances are $R_1 = 0.35R, R_2 = 3R, R_3 = 0.55R$, and $R_4 = 3R$, where $R = 19 \Omega$. The circuit is connected to a $V = 2.7 \text{ V}$ source.

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$$R_4 + \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$



$$P = \frac{V^2}{R_{eq}}$$

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Part (a) Input an expression for the equivalent resistance of the circuit in terms of R_1, R_2, R_3 , and R_4 .

Expression :

$$R_{eq} = \text{_____}$$

Select from the variables below to write your expression. Note that all variables may not be required.

$\alpha, \beta, \theta, a, b, d, g, h, j, k, m, P, R_1, R_2, R_3, R_4, R_{eq}, S, t, V$

Part (b) Input an expression using the R_{eq} and V for power expended by the circuit.

Expression :

$$P = \frac{V^2}{R_{eq}}$$

Select from the variables below to write your expression. Note that all variables may not be required.

$\alpha, \beta, \theta, d, g, h, m, P, R_1, R_2, R_3, R_4, R_{eq}, t, V$

Part (c) What is the power P , in watts?

Numeric : A numeric value is expected and not an expression.

$$P = \text{_____}$$

Problem 5: A current of $I = 4.9 \text{ A}$ passes through the circuit shown, where $R = 39 \Omega$.

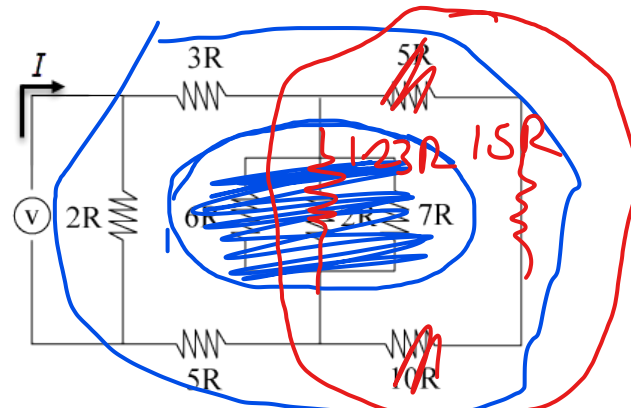
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$$\frac{1}{\frac{1}{6} + \frac{1}{2} + \frac{1}{7}} = 1.2353R$$

$$\frac{1}{\frac{1}{12353} + \frac{1}{15}} = 1.141R$$

$$\frac{1}{\frac{1}{2} + \frac{1}{9.141}} = 1.641R$$

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$$1.141R + 3R + 5R = 9.141R$$

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Part (a) In terms of R , I , and numeric values, write an expression for the voltage of the source, V .

Expression :

$$V = 1.64 I R$$

Select from the variables below to write your expression. Note that all variables may not be required.

$\alpha, \beta, \theta, a, d, g, h, I, j, k, m, P, R, S, t$

Part (b) What is the voltage, V in volts?

Numeric : A numeric value is expected and not an expression.

$$V =$$

Problem 6: Consider the circuit in the diagram, with sources of emf listed below.

Randomized Variables

$$\mathcal{E}_1 = 28 \text{ V}$$

$$\mathcal{E}_2 = 42 \text{ V}$$

$$\mathcal{E}_3 = 8.5 \text{ V}$$

$$\mathcal{E}_4 = 33 \text{ V}$$

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KCL

$$I_3 = I_1 + I_2 \quad \text{--- (1)}$$

KVL for (1)

$$\mathcal{E}_2 - I_2 r_2 - I_2 R_2 + I R_3 + I_1 r_1 - \mathcal{E}_1 + R_1 I_1 = 0$$

$$42 - 40.5 I_2 + 25.1 I_1 - 28 = 0$$

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Part (a) Find I_1 in amps.

Numeric : A numeric value is expected and not an expression.

$$I_1 = -0.115 \text{ A}$$

Part (b) Find I_2 in amps.

Numeric : A numeric value is expected and not an expression.

$$I_2 = 0.2496$$

Part (c) Find I_3 in amps.

Numeric : A numeric value is expected and not an expression.

$$I_3 = 0.0945$$

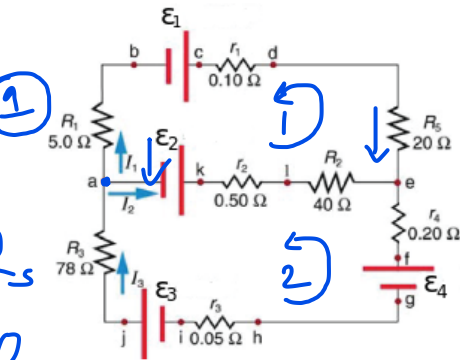
Problem 7: Consider a circuit shown in the figure. Ignore the internal resistances of the batteries.

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$$-I_3 R_3 + \mathcal{E}_1 - R_1 I_1 = 0 \quad \checkmark$$

$$I_3 + I_2 = I_1 \quad \checkmark$$

$$R_1 I_1 - \mathcal{E}_1 - \mathcal{E}_2 + I_2 R_2 = 0$$

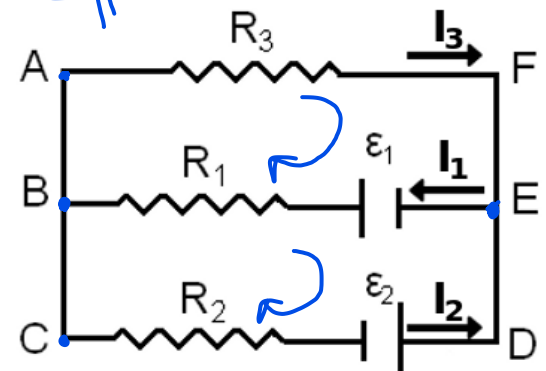
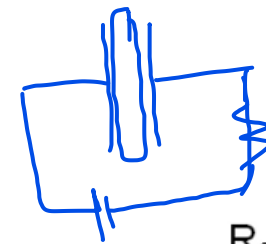


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$$25.1 I_1 - 40.5 I_2 = -14 \quad \text{--- (2)}$$

$$I_3 R_3 - \mathcal{E}_3 + I_3 r_3 + \mathcal{E}_4 + r_4 I_3 + R_2 I_2 + I_2 r_2 - \mathcal{E}_2 = 0$$

$$78.25 I_3 + 40.5 I_2 = 17.5 \quad \text{--- (3)}$$



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Part (a) Express the current I_1 going through resistor R_1 in terms of the currents I_2 and I_3 going through resistors R_2 and R_3 . Use the direction of the currents as specified in the figure.

Expression :

$$I_1 = I_2 + I_3$$

Select from the variables below to write your expression. Note that all variables may not be required.

$\mathcal{E}_1, \mathcal{E}_2, \alpha, \beta, \theta, d, g, I_1, I_2, I_3, m, R_1, R_2, R_3, t$

Part (b) Write the equation of potential change in loop EBAF in terms of the circuit elements.

Expression :

$$\Sigma V_{\text{EBAF}} = -I_3 R_3 + \mathcal{E}_1 - I_1 R_1$$

Select from the variables below to write your expression. Note that all variables may not be required.

$\mathcal{E}_1, \mathcal{E}_2, \alpha, \beta, \theta, d, g, I_1, I_2, I_3, m, R_1, R_2, R_3, t$

Part (c) Write the equation of potential change in loop DCAF in terms of the circuit elements.

Expression :

$$\Sigma V_{\text{DCAF}} = I_1 R_1 - \mathcal{E}_1 - \mathcal{E}_2 + I_3 R_3$$

Select from the variables below to write your expression. Note that all variables may not be required.

$\mathcal{E}_1, \mathcal{E}_2, \alpha, \beta, \theta, d, g, I_1, I_2, I_3, m, R_1, R_2, R_3, t$

Part (d) Solve the three equations to get I_3 .

Expression :

$$I_3 = \frac{\mathcal{E}_2}{2R_3}$$

Select from the variables below to write your expression. Note that all variables may not be required.

$\mathcal{E}_1, \mathcal{E}_2, \alpha, \beta, \theta, d, g, I_1, I_2, I_3, m, R_1, R_2, R_3, t$

Part (e) Now let $\mathcal{E}_1 = 34 \text{ V}$, $\mathcal{E}_2 = 16 \text{ V}$, $R_1 = 1 \Omega$, $R_2 = 1 \Omega$, and $R_3 = 2 \Omega$. Calculate the numerical value of I_3 in amps.

Numeric : A numeric value is expected and not an expression.

$$I_3 = \text{A}$$

Part (f) Calculate the numerical value of I_2 in amps.

Numeric : A numeric value is expected and not an expression.

$$I_2 = \text{A}$$

Part (g) Calculate the numerical value of I_1 in amps.

Numeric : A numeric value is expected and not an expression.

$$I_1 = \text{A}$$

Problem 8: Two capacitors of capacitance $3C$ and $5C$ (where $C = 0.055 \text{ F}$) are connected in series with a resistor of resistance $R = 5.5 \Omega$.

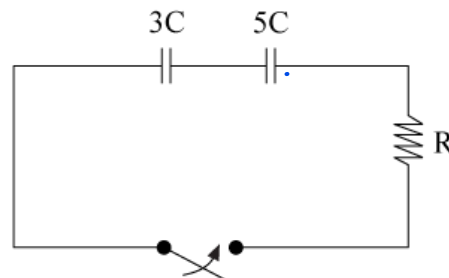
Randomized Variables

$$R = 5.5 \Omega$$

$$C = 0.055 \text{ F}$$

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$$\frac{1}{\frac{1}{3} + \frac{1}{5}} = \frac{15}{8} \times 0.055$$



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Part (a) How long will it take the amount of charge in the circuit to drop by 75% in seconds?

Numeric : A numeric value is expected and not an expression.

$$t = \text{s}$$

Part (b) If the circuit was charged by a 10.0 V source how much total charge (in C) did both capacitors have in them to begin with?

$$Q = CV$$

$$Q_0 = \frac{15}{8} \times 0.055 \times 10$$

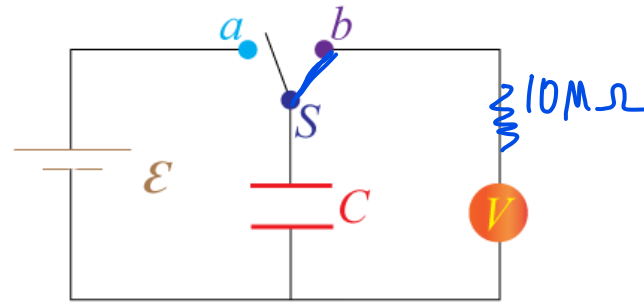
Numeric : A numeric value is expected and not an expression.

$$q_i = \underline{\hspace{2cm}} \text{ C}$$

Problem 9: While preparing a laboratory experiment on capacitors, you connect an $\varepsilon = 7.5 \text{ V}$ DC power supply to fully charge an old capacitor that is marked $C = 220 \mu\text{F}$. Your lab assistant hooks up the circuit as shown below. When the switch S is at point a , the capacitor is connected to the power supply. When the switch S is at point b , the capacitor is disconnected from the power supply and connected to a voltmeter, V . The voltmeter has an internal resistance $R = 10.0 \text{ M}\Omega$.

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$$Q_0 = C\varepsilon = 7.5 \times 220 \times 10^{-6} = 1.65 \text{ mC}$$



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Part (a) The capacitor is initially uncharged, when the switch is moved to position a , hooking the capacitor C to the power supply. What will happen to the charge on the capacitor?

MultipleChoice :

- 1) It will remain unchanged.
- 2) It will very slowly increase until the capacitor is disconnected from the power supply.
- ☒ 3) It will quickly (but not instantaneously) increase to a maximum value.
- 4) It will quickly (but not instantaneously) increase until the capacitor is disconnected from the power supply.
- 5) It will instantaneously increase to a maximum value
- 6) It will very slowly increase, eventually reaching a maximum value.

$$10^1 \times 10^6 = 10^7$$

Part (b) Calculate the charge on the capacitor (in Coulombs) when it is fully charged by the power supply.

Numeric : A numeric value is expected and not an expression.

$$q = \underline{1.65 \times 10^{-3} \text{ C}}$$

$$\ln(2) / 10^7 \times 220 \times 10^{-6}$$

Part (c) After the capacitor is fully charged, you switch the switch to position b at time $t = 0$ seconds. The voltmeter initially reads 7.5 V . After how much elapsed time (in seconds) would you expect the voltmeter to read $1/2$ its initial value?

Numeric : A numeric value is expected and not an expression.

$$t_{1/2} = \underline{3.15 \times 10^{-4} \text{ s}}$$

$$C = \frac{KA\varepsilon_0}{d}$$

Part (d) When you actually move the switch to point b , the voltmeter shows it only takes 1000.0 s for the voltage to drop to half its initial value. Which of the following are possible explanations of the observed behavior of this circuit? Select all that apply.

MultipleSelect :

- ☒ 1) The dielectric in the capacitor has degraded over time, decreasing the effective capacitance of the capacitor.
- 2) The capacitor has been physically damaged, effectively bringing the plates closer together. ☒
- ☒ 3) The internal resistance of the voltmeter is significantly lower than $10 \text{ M}\Omega$.
- 4) The internal resistance of the power supply is acting in series with the capacitor, decreasing the overall resistance of the circuit.
- ☒ 5) The internal resistance of the power supply is acting in parallel with the capacitor, decreasing the overall resistance of the circuit.

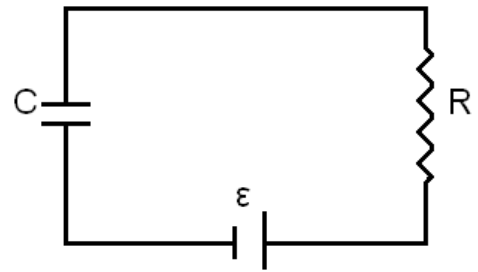
$$\begin{array}{l} \text{Series resistors: } 2 + 2 = 4 \\ \text{Parallel resistors: } \frac{2 \times 2}{2 + 2} = 1 \Omega \end{array}$$

$R_{eq} > 50$
 $R_{eq} < 1$

$\frac{1}{50} \times 4$

Problem 10: An uncharged capacitor with $C = 9 \mu\text{F}$ and a resistor with $R = 95 \Omega$ are connected in series with a battery of $\epsilon = 2.5 \text{ V}$.

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CE

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Part (a) Express the time constant τ in terms of R and C .

Expression :

$\tau = RC$

Select from the variables below to write your expression. Note that all variables may not be required.

$\alpha, \beta, \epsilon, \theta, C, d, g, h, i, j, k, m, P, R, t$

Part (b) Calculate the numerical value of τ in microseconds.

Numeric : A numeric value is expected and not an expression.

$\tau = \frac{1}{1} \mu\text{s}$

Part (c) Express the maximum charge Q on the capacitor in terms of C and ϵ .

Expression :

$Q = CE$

Select from the variables below to write your expression. Note that all variables may not be required.

$\alpha, \beta, \epsilon, \theta, C, d, g, h, i, j, k, m, P, R, t$

Part (d) Calculate the numerical value of Q in microcoulombs.

Numeric : A numeric value is expected and not an expression.

$Q = \frac{1}{1} \mu\text{C}$